

A 0.116 pJ/bit Latch-Based True Random Number Generator Featuring Static Inverter Selection and Noise Enhancement

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Abstract—This article presents a true random number generator (TRNG) that achieves high entropy generation across wide voltage and temperature (VT) range (0.3–1.0 V, -40°C to 110°C) in a single latch-based entropy source (ES). In the ES, static inverter selection technique to minimize the mismatch between the paired inverters, and noise enhancement methods to increase the root mean square (rms) of noise voltage (σ_n) are implemented for good randomness and robustness. In a 130-nm CMOS technology, the TRNG occupies $5343\ \mu\text{m}^2$ and consumes 0.116 pJ/bit at 0.3 V including an on-chip von Neumann post-processing circuit. The cryptographic quality of TRNG's output is verified by National Institute of Standards and Technology (NIST) SP800-22 tests. Up to 325 mV V_{pp} noise injection attack tolerance is confirmed by power supply frequency injection attack. And an equivalent 20-year life at 0.3 V, 25°C is verified by accelerated NBTI aging test.

Index Terms—Aging test, entropy, hardware security, latch, National Institute of Standards and Technology (NIST) tests, post-processing, power supply frequency injection attack, true random number generator (TRNG).

I. INTRODUCTION

TRUE random number generator (TRNG) is a device that generates dynamic entropy from physical stochastic processes. Low energy and high robustness TRNG is crucial in information security applications of edge devices for generating session keys, nonces, and initialization vectors [1]. In addition to supporting hardware security, TRNG is becoming attractive for non-von Neumann computing architecture, such as neuromorphic computing [2], ising machine [3], and stochastic computing [4], in which large amounts of random bits are required.

Among the recent TRNG designs reported in [5], [6], [7], [8], and [9], latch-based TRNGs have demonstrated merits in low energy consumption and small area, owing to their

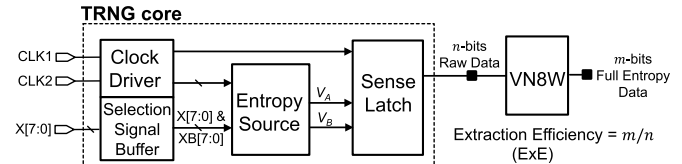


Fig. 1. Block diagram of the proposed TRNG. X[7:0] and XB[7:0] serve as the control signals for mismatch calibration of the ES.

simple cross-coupled structure and the fact that only one-time voltage transition is needed to generate one random bit. However, the mismatch between the paired inverters of a latch introduces bias, resulting in entropy reduction and randomness degeneration. To mitigate this issue, various approaches have been proposed. Satpathy et al. [5] introduce a feedback control circuit for run-time mismatch calibration, but this comes at the cost of increased energy consumption and area. Although Zhang et al. [9] remove the feedback control circuit, it still requires additional capacitors for mismatch compensation and XORs the output of four latch-based entropy sources (ESs) for sufficient and robust raw entropy generation, which limits the further reduction of area and energy consumption.

In this article, based on our work [10], we propose a low-power latch-based TRNG that achieves high and robust raw entropy generation in a single latch-based ES. Static inverter selection technique replaces the additional capacitors [9] for mismatch reduction and merits larger kT/C noise. In addition, noise enhancement technique using damped oscillation mode [9] is applied in this design. An on-chip 8-bit von Neumann with waiting circuit (VN8W) [11] is implemented for the high-efficient raw entropy distillation.

The remainder of this article is structured as follows. Section II provides the circuit structure and operating principle of the proposed TRNG. Section III presents the measurement results. Section IV gives conclusions.

II. CIRCUIT DESIGN

Fig. 1 depicts the block diagram of the proposed TRNG. The raw random bits are generated by the TRNG core, in which the key component is a latch-based ES. The clock driver generates three clocks for controlling switch S_1 , S_2 , and S_3 in the ES, and one sense-enable clock for the sense latch circuit. The selection signal buffer generates the inverter selection signals X[7:0] and XB[7:0] for mismatch calibration of the ES. By manipulating these two sets of signals, the inverters with

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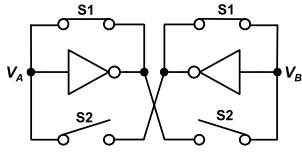


Fig. 2. Simplified structure of latch-based ES.

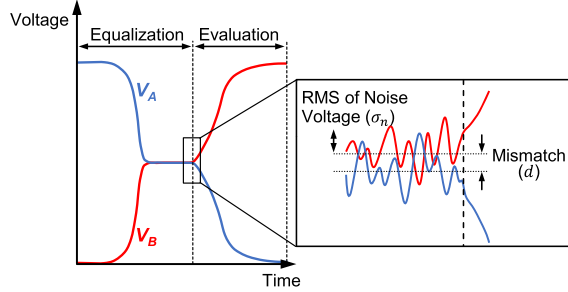


Fig. 3. Voltage waveform of latch-based ES in one cycle.

the minimal mismatch are selected for pairing. Raw random data are latched in the sense latch circuit, which is designed as a strong arm latch [12]. An 8-bit von Neumann with waiting circuit (VN8W) [11] further extracts the entropy from raw data.

A. Latch-Based ES

Fig. 2 is the simplified structure of a latch-based ES. Fig. 3 shows the voltage transition waveform in one typical operation cycle of the ES. During the equalization phase, switch S1 is on, while switch S2 is off, the input and output of inverter are short-connected via switch S1, both inverters are forced to be equalized. Afterward, S1 turns off and S2 turns on, coupling the two inverters to form a latch structure. The differential noise voltage between node “A” and “B” pushes the latch to a random bi-stable state ($A = 1, B = 0$ or $A = 0, B = 1$), resulting in the generation of one random bit. However, the resolution direction towards the bi-stable state is affected not only by noise but also by the mismatch between the paired inverters. Under the combined influence of these two factors, the ES generates a biased random bit stream. The relationship among the root mean square (rms) of noise voltage (σ_n), the mismatch factor (d), and the probability of “1” ($p1$) in the raw random bit stream is modeled as the following equation [13], [14]:

$$\frac{d}{\sigma_n} = \Phi^{-1}(p1) \quad (1)$$

where Φ^{-1} is the reverse function of cumulative distribution function (cdf) of the standard normal distribution $\mathcal{N}(0, 1)$.

For a high-quality ES, $|d/\sigma_n|$ smaller than 0.7 is required [15], where $0.24 < p1 < 0.76$. Furthermore, according to [16], the mismatch to noise variation ratio (σ_d/σ_n) should be less than 0.1. To meet these requirements, the static inverter selection technique to reduce σ_d and the noise enhancement techniques to increase σ_n are implemented in the latch-based ES.

B. Static Inverter Selection

Process variation is categorized into two types: global variation and local variation. Among them, only the local variation is the source of inverter mismatch in the ES. Fig. 4 shows the proposed latch-based ES. Both sides of the latch incorporate four parallel, identically sized inverters (termed as “inverter pack”) that are controlled by the selection signals $X[7:0]$ and their reverse $XB[7:0]$. By manipulating these signals, we can connect an inverter (or multiple inverters) to the common node ac or BC from both sides that exhibit the smallest $|d|$ for pairing. When the $|d|$ is minimized, owing to the fact that σ_n remains consistent, the ratio $|d/\sigma_n|$ achieves its minimum value, thereby yielding the highest entropy. For example, if the $|d|$ between inverters No. 0 and No. 7 is minimal, then we set $X[3:0] = 0001$, $XB[3:0] = 1110$ and $X[7:4] = 1000$, $XB[7:4] = 0111$ to select No. 0 and No. 7 to build a latch. Similarly, any two inverters can be selected from the inverter pack on both sides. There are in total 16 ($C_4^1 \cdot C_4^1$) combinations of 1 inverter by 1 inverter (noted as 1 by 1) and 36 ($C_4^2 \cdot C_4^2$) combinations of two inverters by two inverters (noted as 2 by 2). In this work, the combinations of 3 by 3 and 4 by 4 are discarded because of large energy consumption.

There are two driving mechanisms for the inverter selection signals. For our prototype system, the signals are managed by an off-chip controller. Therefore, we can manually change the inverter selection code to conduct a comprehensive test of TRNG. But, for mass production, the inverter selection signals are sourced from an on-chip non-volatile memory, which stores the signal code that minimizes the $|d/\sigma_n|$ ratio at 0.6 V, 25 °C. This code is written by the manufacturer before chip shipping, and is not modifiable by the customer, which is so-called static selection.

In the following part, the reduction of mismatch by this inverter selection method is estimated. To do this, 1) we construct the statistical model to analyze the distribution of $|d|$ before and after inverter selection process. 2) We quantitatively illustrate the reduction of $|d|$ in terms of the reduction ratio. The calculated reduction ratio is $0.087\times$ and $0.039\times$ for the 1 by 1 and 2 by 2 selection methods, respectively.

- 1) Before selection, the original probability density function (pdf) and cdf of d are denoted by $f(d)$ and $F(d)$. Thus, the pdf and cdf of the absolute value $|d|$, denoted by $f_{\text{abs}}(d)$ and $F_{\text{abs}}(d)$, are given by the following equations:

$$f_{\text{abs}}(x) = f(d) + f(-d) \quad (2)$$

$$F_{\text{abs}}(x) = F(d) - F(-d) \quad (3)$$

where x corresponds to $|d|$, $x \in [0, +\infty)$.

After selection, the pdf noted as $g_{\text{abs}}(x)$ and cdf noted as $G_{\text{abs}}(x)$ of the smallest $|d|$ selected from k identically and independently distributed candidates ($|d_1|, |d_2|, \dots, |d_k|$) are calculated by the following equations:

$$g_{\text{abs}}(x) = k f_{\text{abs}}(x) (1 - F_{\text{abs}}(x))^{k-1} \quad (4)$$

$$G_{\text{abs}}(x) = 1 - (1 - F_{\text{abs}}(x))^k \quad (5)$$

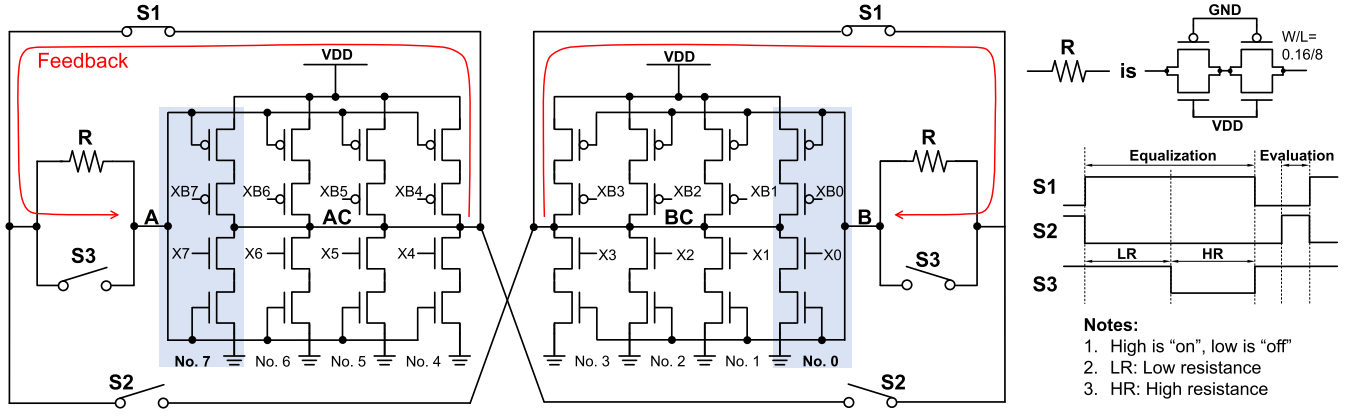


Fig. 4. Circuit of the proposed latch-based ES with static in-cell inverter selection for mismatch calibration, and noise enhancement; implementation of resistance R ; clock diagram in one cycle.

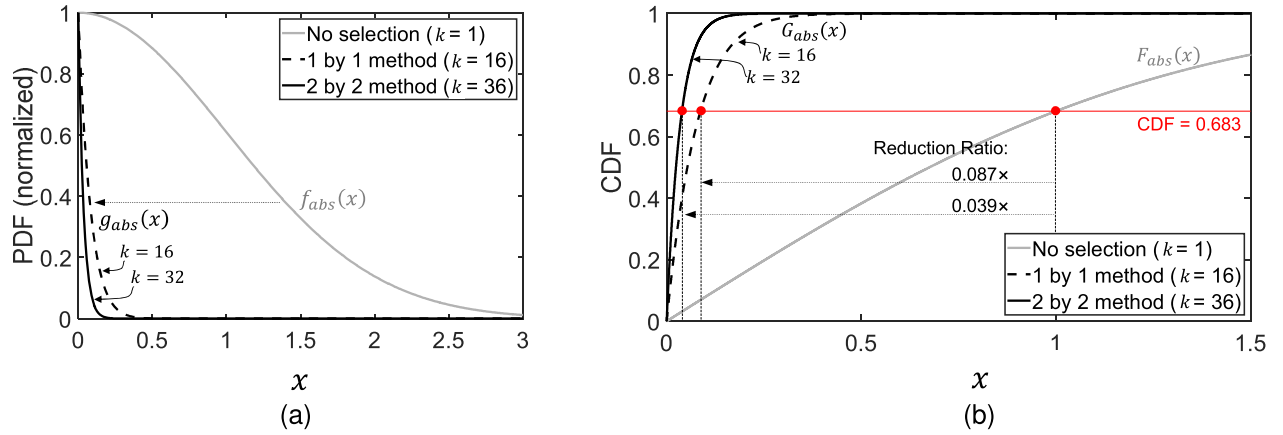


Fig. 5. (a) Probability density function of $|d|$ before and after inverter selection. (b) Cumulative distribution function of $|d|$ before and after inverter selection. The reduction ratio is $0.087\times$ and $0.039\times$ for the 1 by 1 ($k = 16$) and 2 by 2 ($k = 36$) selection method, respectively.

In (4), k stands for the combination to choose the smallest one from k candidates. $f_{\text{abs}}(x)$ stands for the probability that the chosen one equals x . $(1 - F_{\text{abs}}(x))^{k-1}$ stands for all others that are larger than x . In (5), $(1 - F_{\text{abs}}(x))^k$ stands for all k candidates that are larger than x , thus, $1 - (1 - F_{\text{abs}}(x))^k$ stands for the smallest one that is smaller than x . Furthermore, the derivative of $G_{\text{abs}}(x)$ is equal to $g_{\text{abs}}(x)$, which meets the general relationship between cdf and pdf.

Fig. 5(a) shows the pdf of $|d|$ before and after inverter selection. In this figure, due to local variation, the original $f(d)$ is assumed to follow a normal distribution $\mathcal{N}(0, 1)$; therefore, the distribution of $f_{\text{abs}}(x)$ conforms to a folded normal distribution, depicted by the gray curve. Afterwards, the 1 by 1 ($k = 16$) or the 2 by 2 ($k = 36$) selection method is applied, the pdf distribution of $|d|$ is compressed in both methods. (In Fig. 5(a), the y-axis values are normalized by their peak value). Fig. 5(b) shows the calculated results of the cdf of $|d|$.

- 2) The reduction ratio is defined as the rate of x values for $\text{cdf} = 0.683$, which correspond to the probability within $\pm 1\sigma$ in the normal distribution. Fig. 5(b) shows the cdf of $|d|$ before and after inverter selection. Before selection, the x value of $F_{\text{abs}}(x) = 0.683$ is 1. After 1 by 1 or 2 by 2 selection, the x values of $G_{\text{abs}}(x) = 0.683$ are

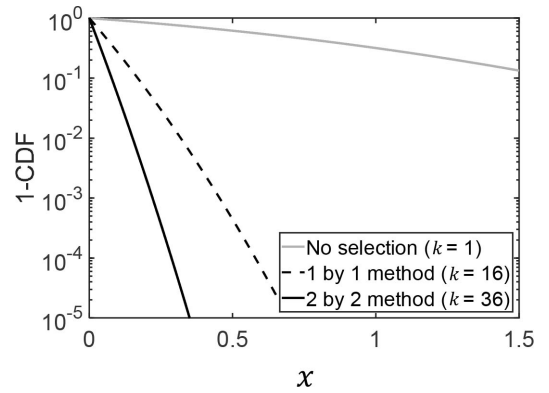


Fig. 6. Probability larger than x ($1 - \text{cdf}(x)$) on a logarithmic scale.

0.087 ($k = 16$) and 0.039 ($k = 36$), respectively. Thus, the reduction ratios are 0.087 for 1 by 1 and 0.039 for 2 by 2 selections. It should be noted that these ratios do not represent absolute mismatch values, and the mismatch before selection in the 2 by 2 method is smaller than that of the 1 by 1 method due to the averaging effect of the two inverters. To illustrate the effect of the selection methods on reducing the probability of having a large mismatch, we present $1 - \text{cdf}(x)$ (the probability larger than x) on a logarithmic scale in Fig. 6.

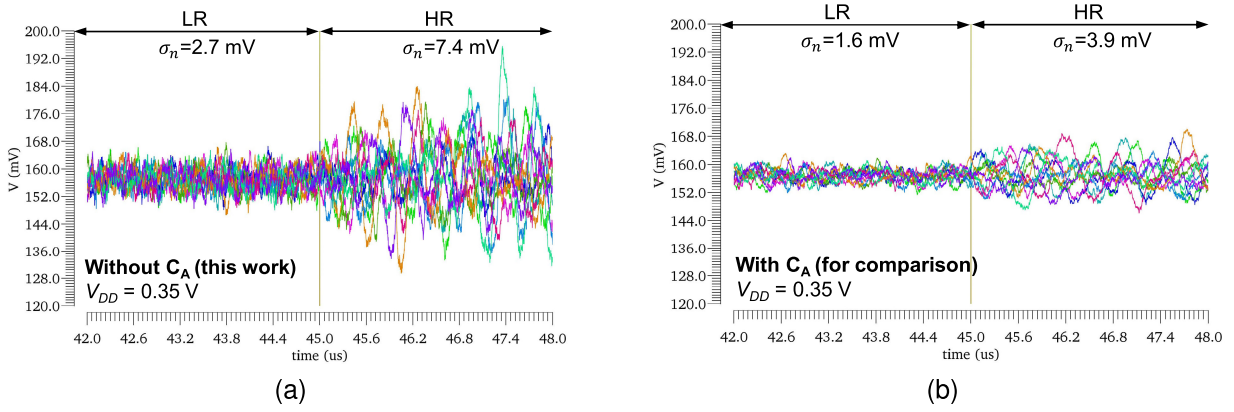


Fig. 7. Monte Carlo noise simulation results of latch-based ES in equalization phase. (a) Without C_A . (b) With C_A . The noise amplification from LR to HR phase, and larger noise at both LR and HR phases in (a) are observed.

C. Noise Enhancement

The noise enhancement technique using damped oscillation mode [9] is applied in this design. A resistor R is inserted into the feedback loop of both side inverters as shown in Fig. 4. $S3$ switches the feedback loop between low resistance (LR) and high resistance (HR) phases. At the beginning of equalization phase, $S1$ and $S3$ are on, $S2$ is off, both side inverters are quickly equalized through the $S1$, $S2$ LR path. Then, $S3$ turns off; the damped oscillation is induced through $S1$, R , and inverter loop, enhancing the noise. In addition, a larger noise voltage than [9] can be obtained in this design. In [9], capacitors are added to the gate input of the inverters to store the voltages in the equalization state. By increasing the capacitance value, the equalized voltages of both sides are closer to the metastable point, thereby compensating for the mismatch. In this work, on the other hand, we do not apply this capacitor-based method. The inverter selection technique reduces the mismatch more effectively and without requiring additional capacitors. Consequently, this leads to smaller node capacitance and larger kT/C noise. Moreover, this results in a smaller area. Although this work has a larger number of MOSFETs, the total TRNG core area is reduced to 66.7%, compared with [9]. Fig. 7 presents the simulated voltage waveform at node “A” without and with C_A (C_A is set to twice the gate capacitance of the inverter pack). During the LR phase, due to σ_n at node “A” is proportional to the square root of kT/C , this design (without C_A) has $1.7\times$ ($2.7/1.6$) larger σ_n compared with the design with C_A [9]. Moreover, during the HR phase, this design shows $1.9\times$ ($7.4/3.9$) larger σ_n compared with the design with C_A [9].

D. von Neumann Post-Processing Circuit

To ensure cryptographic-quality randomness, post-processing of the raw data generated by the latch-based ES is necessary. In this work, the 8-bit von Neumann circuit (VN8W) [11] is implemented for post-processing, which is an upgraded version of conventional von Neumann (2-bit) debiasing and 4-bit von Neumann circuit [17]. VN8W has a higher extraction efficiency (ExE), which is defined as m/n , where m is the number of bits generated by the post-processing circuit and n is the number of raw bits.

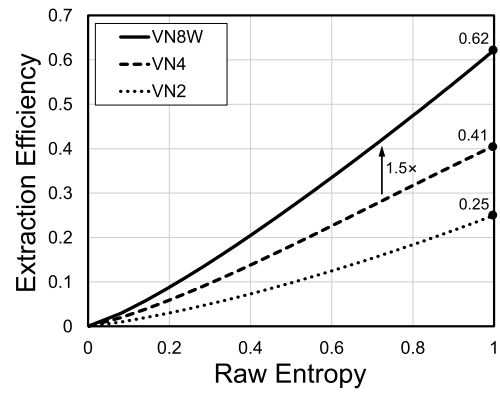


Fig. 8. ExE of VN8W, VN4, and the conventional 2-bit von Neumann circuit (VN2) versus the raw entropy (the Shannon entropy of raw data).

Fig. 8 shows the ExE versus the Shannon entropy of raw bits (raw entropy). The ExE of VN8W is improved by $1.5\times$ compared with VN4. Furthermore, since the post-processing energy consumption per extracted bit of VN8W and VN4 is similar, the total TRNG energy (TRNG core energy/ExE + post-processing energy) is lower by implementing VN8W.

III. MEASUREMENT RESULTS

The proposed TRNG is fabricated in a 130-nm CMOS technology. The chip micrograph is shown in Fig. 9. TRNG core occupies only $443 \mu\text{m}^2$, with VN8W post-processing circuit of $4900 \mu\text{m}^2$; the total area of TRNG is $5343 \mu\text{m}^2$, which is equivalent to $0.316 \times 10^6 F^2$.

Fig. 10 shows the measurement setup. A controller is for data logging and inverter selection signal setting. The arbitrary function generator (AFG) is for clock generation and noise injection attack. The temperature of DUT in the chamber is monitored with a thermocouple wire.

A. Performance of Inverter Selection

Ten chips were measured at 0.6 V supply voltage, 25°C . For each chip's TRNG core, $p1$ for all 16 combinations of 1 by 1 and 36 combinations of 2 by 2 are measured. Then, d/σ_n for each combination is calculated based on (1).

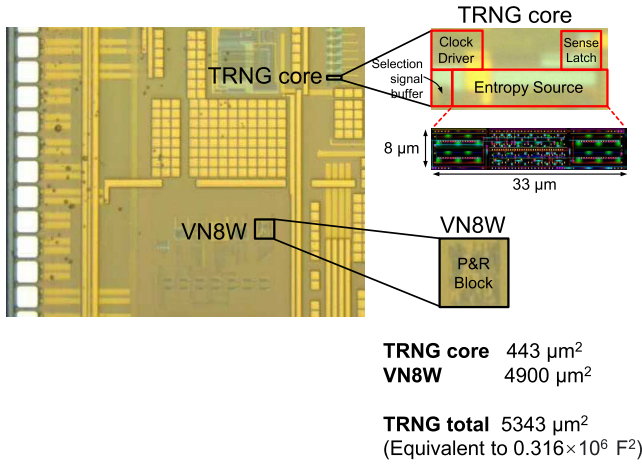


Fig. 9. Chip micrograph and the layout of ES.

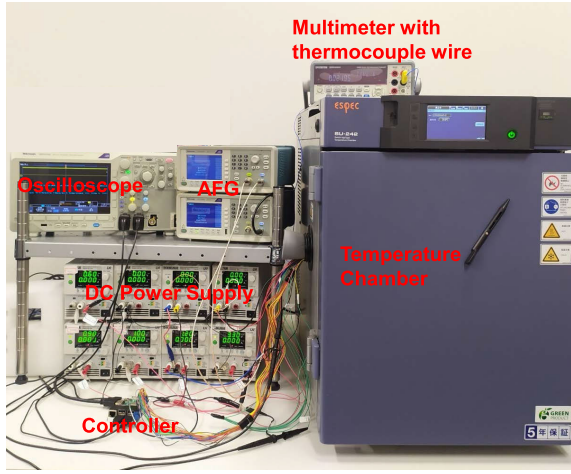


Fig. 10. Measurement setup.

Fig. 11(a) and (b) shows the distribution of d/σ_n of ten chips. When the inverter selection is not applied, the distribution of d/σ_n is shown in gray, and the standard deviation (σ) of d/σ_n is 2.115 (1 by 1) and 1.066 (2 by 2). The smaller σ in 2 by 2 is owing to the smaller mismatch (d) brought by the double-sized MOSFET channel width.

When the inverter selection is applied, the distribution of d/σ_n is dramatically compressed as illustrated in red. In 1 by 1 and 2 by 2 inverter selection methods, the mismatch to noise variation ratio (σ_d/σ_n), which is the σ of d/σ_n , is reduced to 0.147 and 0.043, respectively. The 2 by 2 selection result meets the target of ($\sigma_d/\sigma_n < 0.1$). The reduction rates of $0.070\times$ and $0.040\times$ meet well with the results of statistical analysis in Section II-B.

When we consider 6σ process variation ($|\mu \pm 6\sigma|$ of d/σ_n), in 1 by 1 selection method, $|\mu \pm 6\sigma| \leq 0.938$, in 2 by 2 selection method, $|\mu \pm 6\sigma| \leq 0.273$. The 2 by 2 selection method still meets the requirement of ($|d/\sigma_n| < 0.7$).

Moreover, Fig. 11(c) shows the distribution of the top three smallest $|d/\sigma_n|$ of each chip in 2 by 2 method. Even if we loosen the selection criteria and arbitrarily choose one combination from the top three for each chip, the resulting

TABLE I
NIST SP800-22 TEST RESULTS

NIST SP800-22 Tests		$V_{DD} = 0.3 \text{ V}$		$V_{DD} = 0.6 \text{ V}$		$V_{DD} = 1.0 \text{ V}$	
		$P\text{-value}^a$	Passed	$P\text{-value}^a$	Passed	$P\text{-value}^a$	Passed
1	Frequency	0.580	16/16	0.574	16/16	0.434	16/16
2	Block Frequency	0.495	16/16	0.475	16/16	0.423	16/16
3	Cumulative Sums	0.550	16/16	0.516	16/16	0.432	16/16
4	Runs	0.543	16/16	0.478	16/16	0.548	16/16
5	Longest Runs	0.422	16/16	0.455	16/16	0.556	16/16
6	Rank	0.454	16/16	0.642	16/16	0.464	16/16
7	FFT	0.462	16/16	0.339	16/16	0.598	16/16
8	Non-overlapping Template	0.502	16/16	0.508	16/16	0.499	16/16
9	Overlapping Template	0.487	16/16	0.535	16/16	0.525	16/16
10	Universal	0.529	16/16	0.560	16/16	0.560	16/16
11	Approximate Entropy	0.553	16/16	0.599	16/16	0.390	16/16
12	Random Excursion	0.506	12/12	0.548	10/10	0.524	11/11
13	Random Excursions Variant	0.500	12/12	0.551	10/10	0.524	11/11
14	Serial	0.461	16/16	0.484	16/16	0.555	16/16
15	Linear Complexity	0.529	16/16	0.596	16/16	0.410	16/16

(a) The averaged $P\text{-Value}$.

σ_d/σ_n value is 0.104, which remains close to the target of ($\sigma_d/\sigma_n < 0.1$).

B. Entropy Generation Against VT Variations

To confirm the performance of static selection, in which run-time reselection is not allowed, the robustness of entropy generation of the fixed selected pair against supply voltage and temperature (VT) variations is tested. As for the V_{DD} dependence, as shown in Fig. 12(a), the average Shannon entropy of raw data, obtained from three different chips, remains stable and consistently exceeds 0.95 across a wide voltage range (0.3–1.0 V). As for temperature dependence, as shown in Fig. 12(b), the average entropy of three chips at the supply voltage of 0.3 and 0.6 V is larger than 0.94 across temperature range -40°C to 110°C . Even the worst raw entropy of the three chips is still above 0.9 in this temperature range.

C. Energy Consumption, Throughput, and Randomness Tests

Fig. 13(a) shows the supply voltage dependence of TRNG core current and cycle time. The minimum current of 0.913 nA with 100 μs cycle time is obtained at 0.3 V supply voltage. Fig. 13(b) shows the energy consumption of the whole TRNG circuit (TRNG core + VN8W) and corresponding throughput versus supply voltage. Considering the ExE of VN8W, the proposed TRNG achieves a minimum energy consumption of 0.116 pJ/bit at 0.3 V (TRNG core: 0.044 pJ/bit + VN8W: 0.072 pJ/bit). Compared with the work [9], as shown in Fig. 13(c), the energy consumption is reduced by a total of 38% and TRNG core of 61%. This energy reduction benefits from that only one ES is required in the proposed TRNG core, while [9] needs XOR the output of four ES for sufficient and robust raw entropy generation. VN8W consumes more than 62% of total energy. By using advance fabrication process technology, this part can be further reduced.

The post-processed data from four chips, and four streams of 1 M bits for each chip, in total 16 streams, have been evaluated by the National Institute of Standards and Technology (NIST) SP800-22 [18] ($P\text{-value} \geq 0.01$) tests. The results shown in Table I confirm the cryptographic-quality randomness.

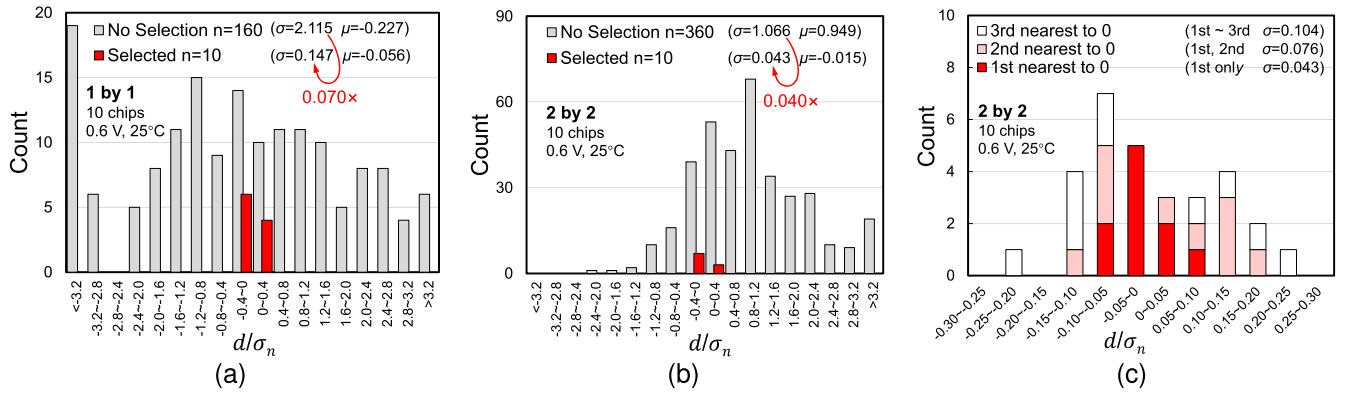


Fig. 11. Measured distribution of d/σ_n . (a) Applying 1 by 1 selection method. (b) Applying 2 by 2 selection method. (c) Distribution of top three smallest d/σ_n in 2 by 2 selection method.

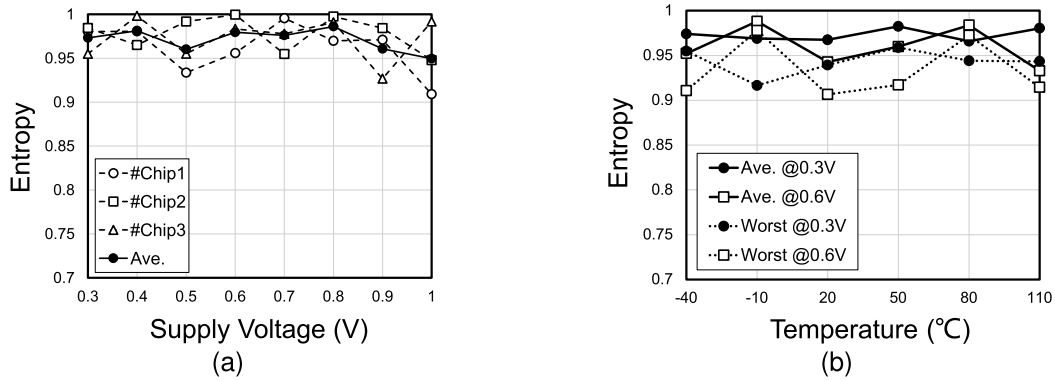


Fig. 12. Raw data entropy: (a) under supply voltage variations and (b) under temperature variations.

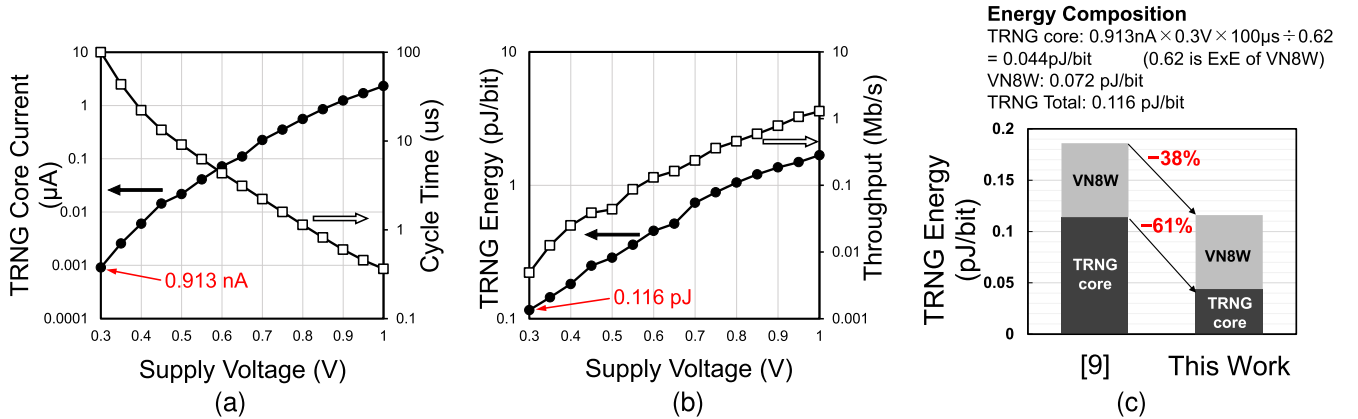


Fig. 13. (a) Supply current and cycle time of TRNG core. (b) Energy consumption and throughput of TRNG. (c) Energy composition and comparison.

D. Supply Noise Injection Attack

The tolerance to power supply noise injection attack [6], [19] is evaluated. Fig. 14(a) shows the Shannon entropy of raw random data under supply noise frequency 0.1–60 MHz with 200 mV V_{pp} , which is the peak to peak voltage of injected sinusoidal noise wave. Several entropy degenerations are observed, and the most severe degradation peak occurs at 23.5 MHz. At this peak condition, the post-processed data passed all NIST SP800-22 tests. Furthermore, Fig. 14(b) shows the noise V_{pp} sweep at this frequency; the Shannon entropy of raw data decreases as V_{pp} increases. Even at 325 mV V_{pp} with as low as 0.64 Shannon entropy, it passed all

NIST SP800-22 tests after VN8W post-processing. A 10-k bit speckle diagram at this pass boundary is shown in Fig. 15, and no specific bit pattern can be found. The fact that the ExE of VN8W exhibits a positive correlation with the entropy of input random bits (Fig. 8), suggests that ExE can be utilized to detect power noise attacks with adequate entropy guard bands. For instance, when we set the ExE lower bound to 0.48 (entropy > 0.8), an entropy guard band is 0.15.

E. Long-Term Reliability

To verify long-term reliability, an accelerated NBTI aging test is applied to a TRNG chip by maintaining the supply

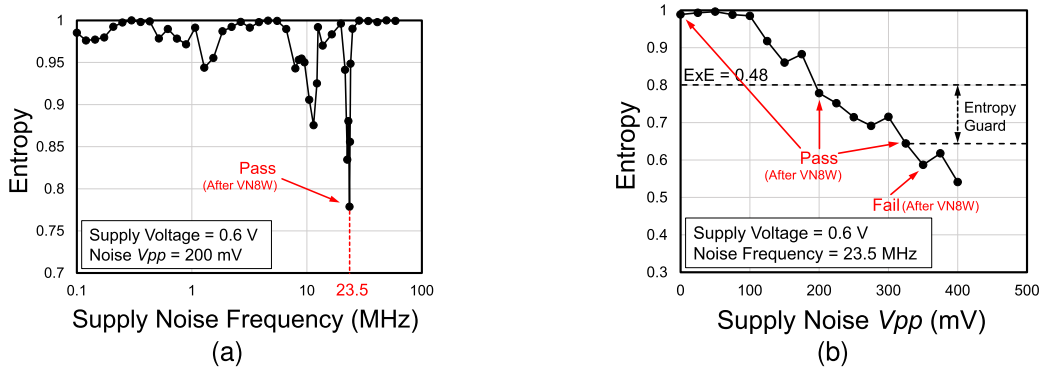
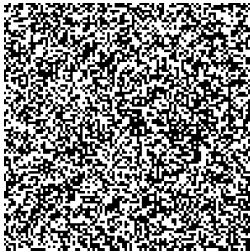


Fig. 14. Supply noise injection attack results. (a) Impacts of supply noise frequency on the Shannon entropy of raw data. (b) Impacts of supply noise amplitude on the Shannon entropy of raw data.

TABLE II
PERFORMANCE COMPARISON

	This Work	JSSC' 22 [9]	JSSC' 19 [5]	TCA-I' 22 [23] ^d	ISSCC' 21 [8]	JSSC' 17 [7]	JSSC' 16 [6]	TVLSI' 21 [24] ^d	TCA-I' 21 [25] ^d
Implementation	ASIC	ASIC	ASIC	FPGA	ASIC	ASIC	ASIC	FPGA	FPGA
Technology (nm)	130	130	14	45	28	180	40	40	40
Structure	Latch	Latch	Latch	Latched-XOR	SRAM	SAR ADC	RO	RO	Jitter-Latch
Entropy Source	Metastability	Metastability	Metastability	Metastability	Leakage noise jitter	Chaotic map	Jitter	Jitter	Jitter
Supply Voltage (V)	0.3~1.0	0.3~1.0	0.55~0.75	1.08~1.32	0.8~1.0	0.6~0.9	0.6~0.9	0.8~1.2	0.9~1.1
Normalized Area ($\times 10^6$ F ²)	0.026 ^a 0.316 ^b	0.039 ^a 0.329 ^b	10.786 ^c	N/A	0.016 ^c	0.14	0.52	N/A	N/A
Min Energy (pJ/bit)	0.116	0.186	2.5	N/A	9.6	0.3	11	N/A	N/A
Max Throughput (Mbps)	1.3	2.39	1480	12.5	3.6	0.27	2	5	100
Temperature (°C)	-40~110	-20~100	25~110	25~75	-10~75	N/A	-40~120	0~80	0~80
Noise Injection Attack	Yes (Up to 325 mV)	Yes	Yes	N/A	N/A	N/A	Yes	N/A	N/A
Long-Term Reliability	Yes	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A

- (a) TRNG Core area
(b) Area includes TRNG core and post-processing circuit.
(c) Post-processing circuit is not included.
(d) These TRNGs have been verified on FPGA and are suitable to be implemented in ASIC.



Condition:
Supply Voltage = 0.6 V
Noise Fre. = 23.5 MHz
Vpp = 325 mV
10k bits (after VN8W)

Fig. 15. Speckle diagram of TRNG data under power supply noise injection attack.

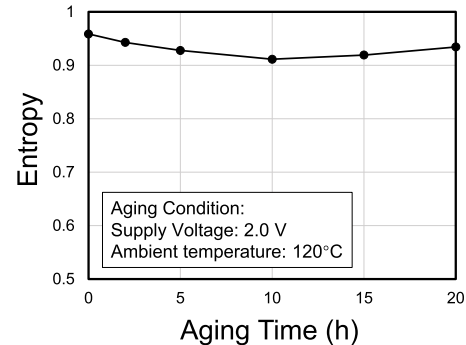


Fig. 16. Accelerated NBTI aging test result.

voltage at 2.0 V, the ambient temperature at 120 °C, and enabling all clocks. Under these conditions, the thermal acceleration factor (TAF) and the voltage acceleration factor (VAF) are 110.1 and 83.1 relative to the operation condition of 0.3 V/25 °C [20], [21]. During a 20-h aging, the Shannon entropy of raw data did not show significant degeneration (Fig. 16). And every data point depicted on this graph passed the NIST SP800-22 tests (pass ratio: 12/12, in 10k bits length) after VN8W post-processing. This indicates an equivalent life of approximately 20 years under the operating condition

of 0.3 V/25 °C ($20 \text{ h} \times 110.1 \times 83.1 = 182986.2 \text{ h} \approx 20.9 \text{ years}$).

F. Comparison

Fig. 17 compares the TRNGs in terms of core area (without post-processing circuit) and energy, and Table II summarizes the measured performance of the proposed TRNG is

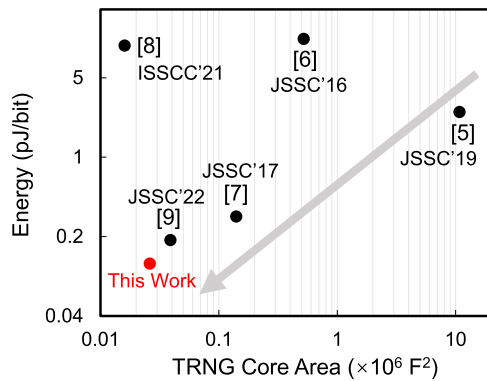


Fig. 17. TRNGs comparison in terms of core area and energy.

comparison to prior works. The proposed TRNG exhibits the smallest energy consumption and the smallest area compared with other latch-based TRNGs [5], [9]. In addition, the robustness against supply noise injection attack and an equivalent 20-year life are confirmed.

IV. CONCLUSION

This article demonstrates a low power and high robustness latch-based TRNG, which achieves dynamic entropy generation in a single latch-based ES across wide voltage and temperature range (0.3–1.0 V, -40°C to 110°C). The static inverter selection technique to minimize the mismatch between the paired inverters, and the noise enhancement methods to increase the rms of noise voltage (σ_n) are implemented for good randomness and robustness. An on-chip von Neumann post-processing circuit (VN8W) is utilized for maximum entropy harvesting. In a 130-nm CMOS technology, the TRNG occupies $5343\ \mu\text{m}^2$ and consumes 0.116 pJ/bit at 0.3 V. The cryptographic quality of TRNG's output is verified by NIST SP800-22 tests. Moreover, up to 325 mV V_{pp} noise injection attack tolerance is tested by power supply frequency injection attack and an equivalent 20-year life at 0.3 V, 25°C is verified by accelerated aging test.

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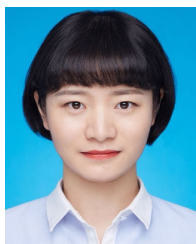
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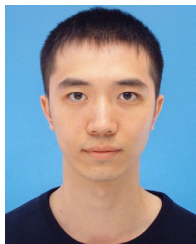
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